

Cloud Computing DevOps (CIS3032-N)

Individual Software Development Plan

Student Details

Student Name:	Ali Mustafa	Student Id:	2110097
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Software Containers to be Developed

Container Name:	Product Service
Descriptor:	The Product Service is a backend microservice responsible for managing the product catalogue for the ThreeAmigos web application. It supports product browsing, filtering, and loose search for public and registered users. The service retrieves product data from a local database and periodically updates pricing and stock information from approved third-party suppliers (mocked in this implementation). It exposes REST APIs consumed by the frontend and validates access tokens for protected endpoints.
System Requirements:	This container supports public product browsing and filtering, loose searching by name and description, and displaying stock availability to registered users. It also contributes to the automatic daily update of product catalogue and pricing, ensuring products are listed using the cheapest supplier price plus a 10% margin.
Technologies:	The Product Service is implemented using Node.js and Express.js and deployed as a Docker container. PostgreSQL is used for persistent storage, while Redis is used as a caching layer to improve performance. Communication is via RESTful APIs over HTTPS.
Data Requirements:	Product, price, and stock data are retrieved from mocked supplier APIs via outbound HTTP calls and stored in a PostgreSQL database. Frequently accessed product listings are cached in Redis to reduce database load and improve resilience during temporary failures.
Endpoints	The service exposes endpoints to retrieve all products, perform loose searches, fetch individual product details, and trigger a daily catalogue refresh. All endpoints are accessed using HTTPS and follow REST conventions.

Provide additional tables for each container you will develop

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Software Containers to be Developed

Container Name:	User Service
Descriptor:	The User Service is a backend microservice responsible for managing registered customer accounts and profiles. It integrates with Auth0 to provide secure authentication and authorisation using OAuth2 and OpenID Connect. The service allows users to register, view and update their profile information, and request account deletion. It also supports staff access using role-based permissions.
System Requirements:	This container supports user registration and secure sign-in, profile management, account deletion requests, and staff access to customer profiles. It contributes to the system's security requirements by enforcing authentication and authorisation for protected operations.
Technologies:	The User Service is built using Node.js and Express.js and deployed as a Docker container. PostgreSQL is used to store user profile data. Authentication and token management are handled externally by Auth0. All communication uses REST over HTTPS.
Data Requirements:	User profile data such as name, address, and telephone number is stored in PostgreSQL. Authentication credentials are not stored locally and are managed by Auth0. Role and account status information is stored to support access control.
Endpoints	The service exposes endpoints for user registration, profile retrieval and update, account deletion requests, and staff-only access to user accounts. All endpoints require HTTPS and validate Auth0-issued JWTs where applicable.

Provide additional tables for each container you will develop